Introduction to Machine Learning

and Introduction to the AI 221 Course

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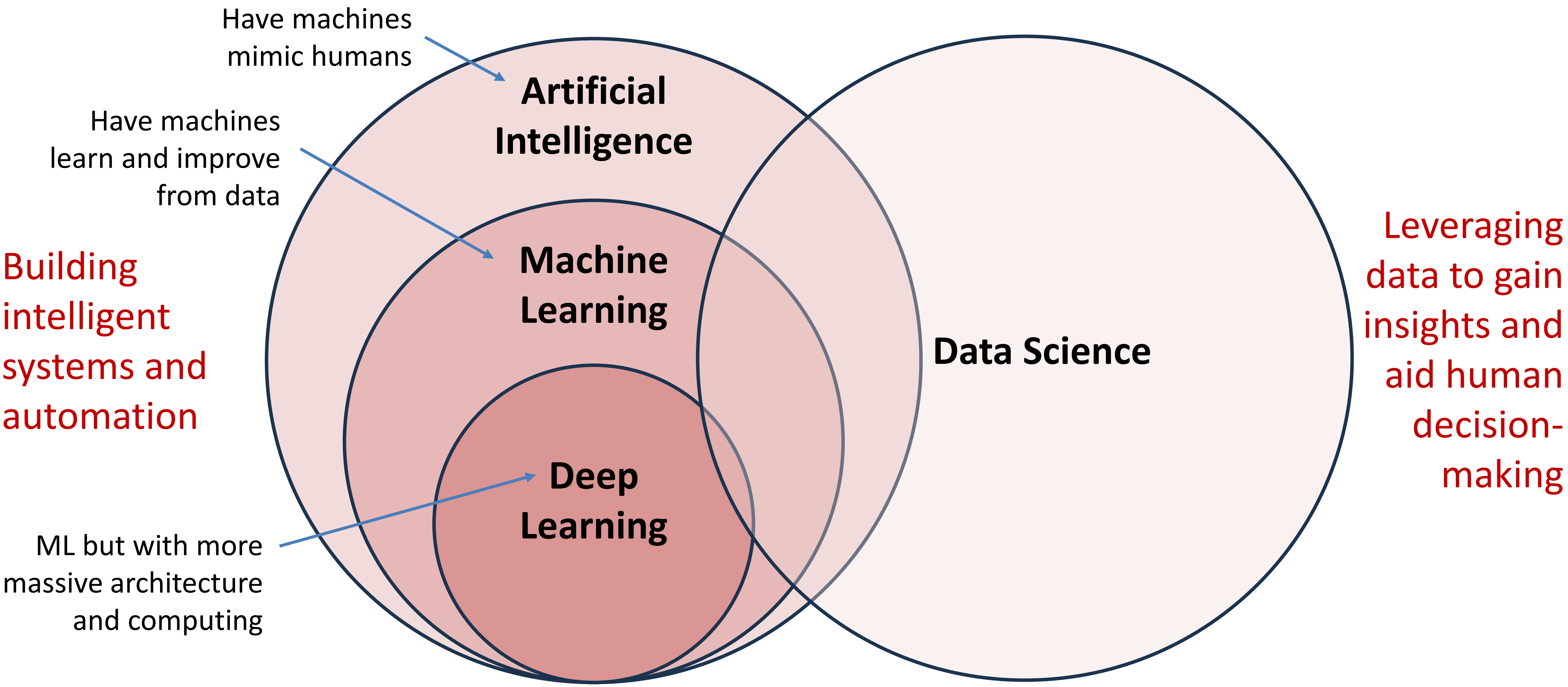
Outline

- What is Machine Learning?
  - Why only now?
  - Why use ML in your industry?
  - How to turn data into decisions?
  - Will machines replace us?
  - Types of Learning Problems

- Intro to the Course (AI 221)
  - Course Delivery
  - Course Content
  - Course Requirements
  - Software

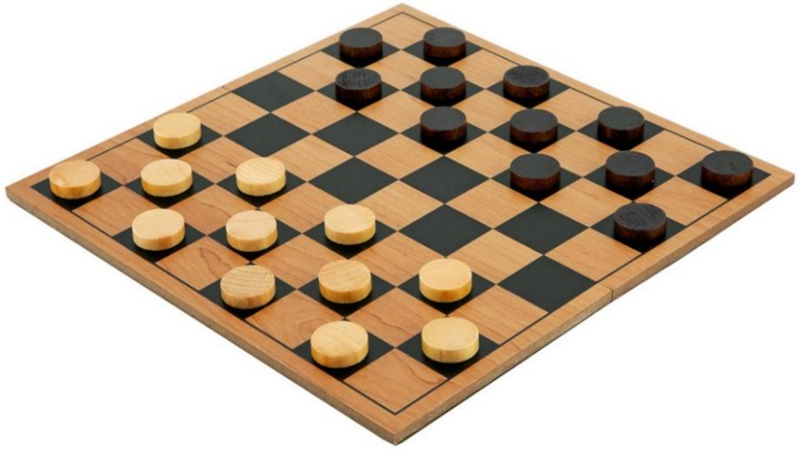
What is AI / ML / DS?



What is Machine Learning?

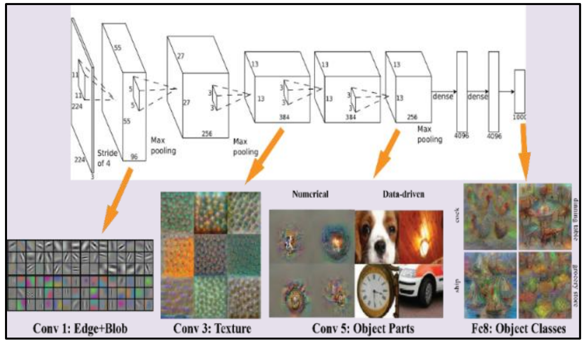
A field of study concerned with giving computers the *ability to learn* without being explicitly programmed.
(Arthur Samuel, 1959)

Arthur Samuel and the IBM 701 Computer



- Arthur Samuel was not a very good checkers player.
- But he was able to program a checkers bot that plays *better than a human*.
- The bot learned by playing millions of times against itself.
- In the process, it learns which moves lead to wins and which moves lead to losses.

IBM Watson
Jeopardy, **2011**

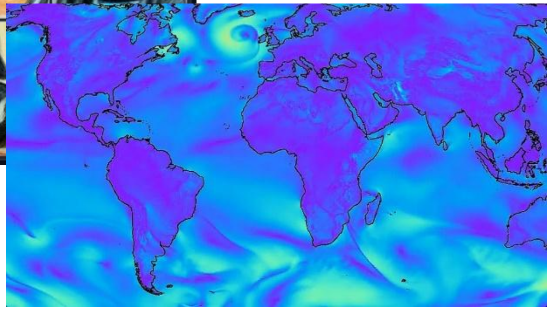


AlexNet
ImageNet Visual Recognition Challenge, **2012**

AlphaGo
Game of Go, **2016**

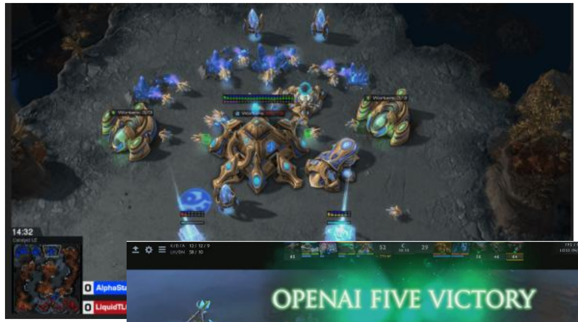
DALL-E
2021, 2022

GraphCast
2023

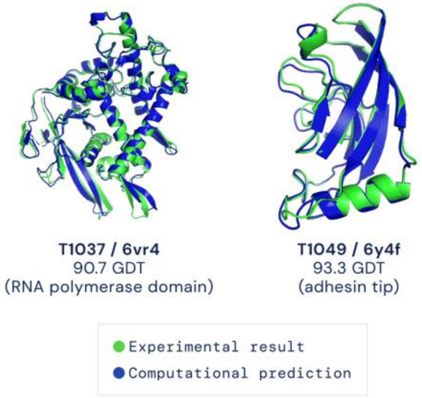


IBM Deep Blue
Chess, **1997**

AlphaStar
StarCraft II, **2019**

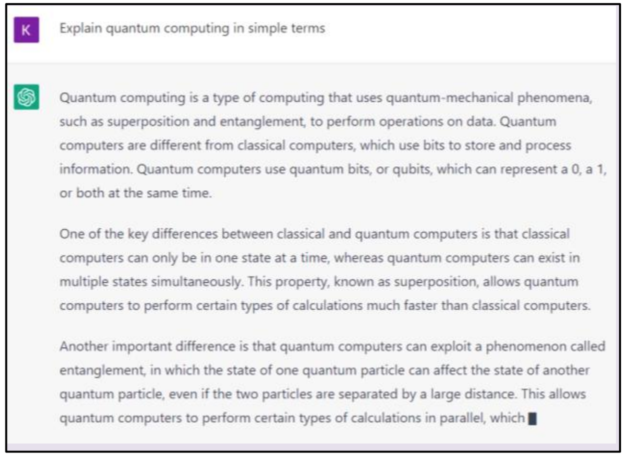


OpenAI Five
Dota 2, **2019**



AlphaFold
Protein Structure Prediction, **2016, 2018**

ChatGPT
2022


Explain quantum computing in simple terms

Quantum computing is a type of computing that uses quantum-mechanical phenomena, such as superposition and entanglement, to perform operations on data. Quantum computers are different from classical computers, which use bits to store and process information. Quantum computers use quantum bits, or qubits, which can represent a 0, a 1, or both at the same time.

One of the key differences between classical and quantum computers is that classical computers can only be in one state at a time, whereas quantum computers can exist in multiple states simultaneously. This property, known as superposition, allows quantum computers to perform certain types of calculations much faster than classical computers.

Another important difference is that quantum computers can exploit a phenomenon called entanglement, in which the state of one quantum particle can affect the state of another quantum particle, even if the two particles are separated by a large distance. This allows quantum computers to perform certain types of calculations in parallel, which

Discriminative vs. Generative A.I.

Since the emergence of Large Language Models and AI-generated images, there is now a distinction between discriminative and generative AI (GenAI).

Discriminative AI

- Predicting, Classifying, Naming
- e.g. Forecast weather, detect faces

CAT

DOG

Generative AI

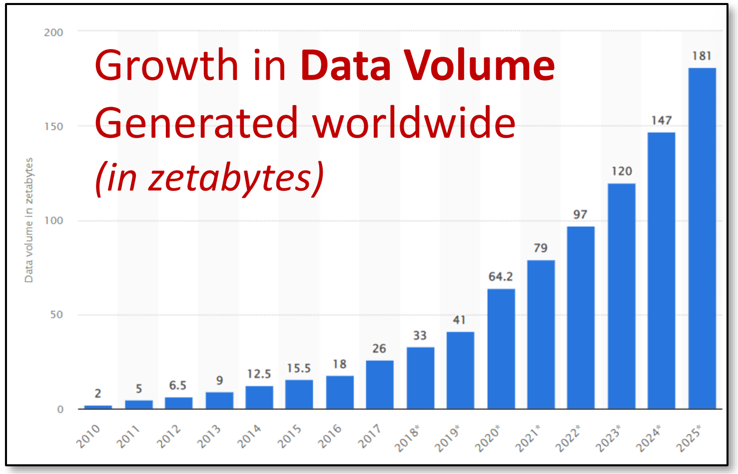
- Sampling, Exploring, Suggesting
- e.g. Generate images, generate text

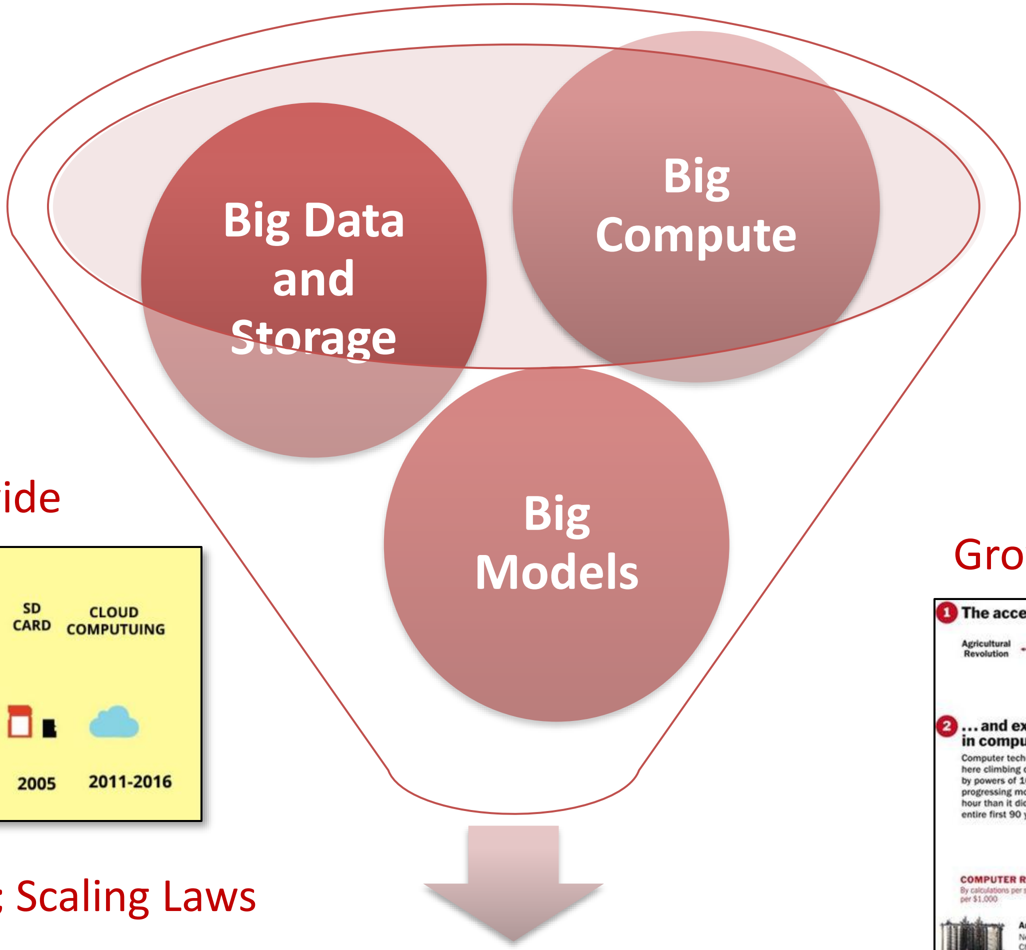
CAT

DOG

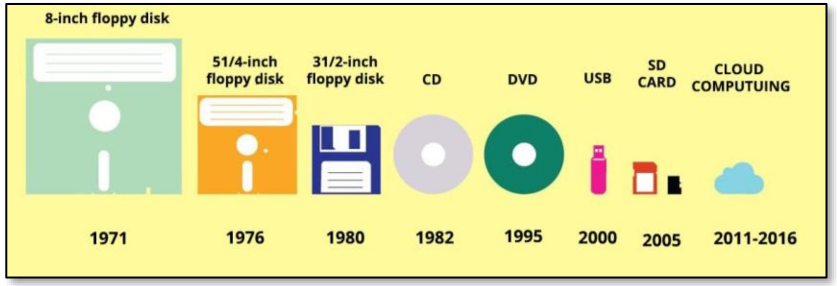
Machine Learning, Data Science, Data Analytics,

…why only now?

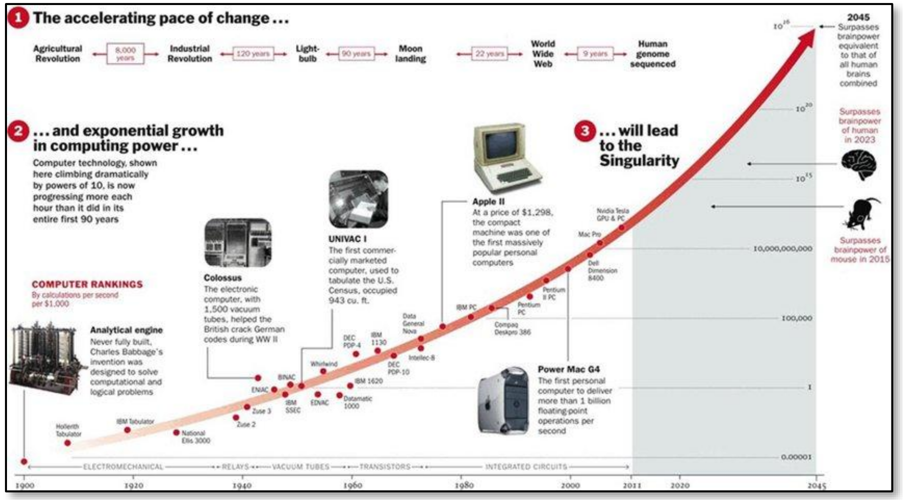




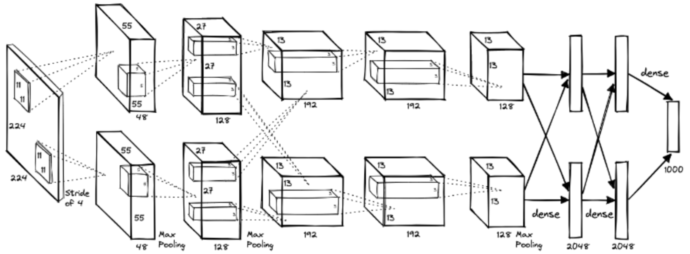
Growth in **Data Storage** worldwide



Growth in **Computing Power** worldwide



Growth in **Model Architectures**; Scaling Laws

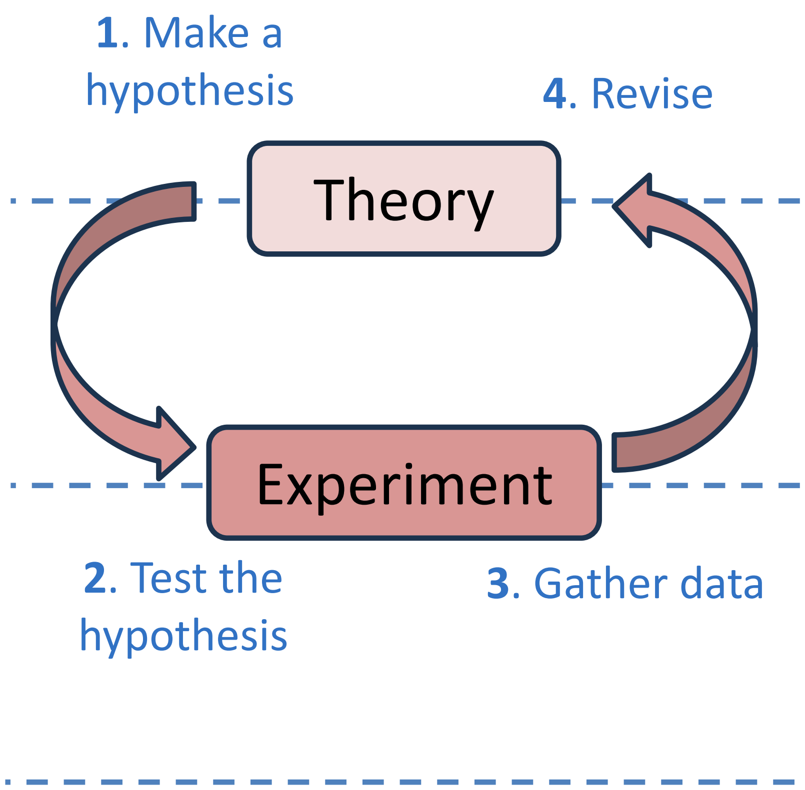


Machine Learning +

Practical Applications

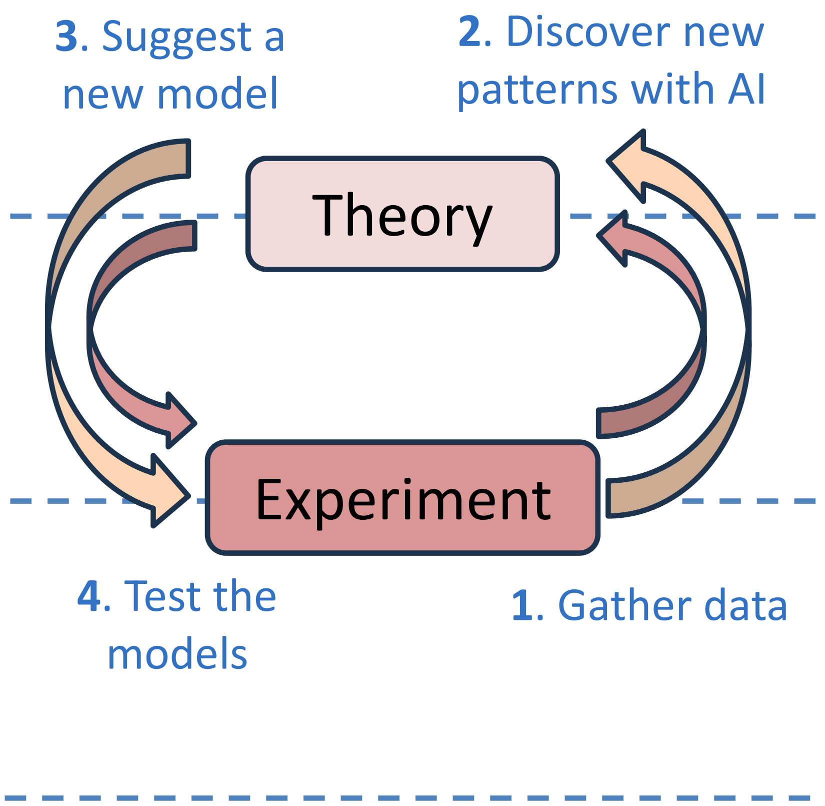
Why should Scientists and Engineers care about AI?

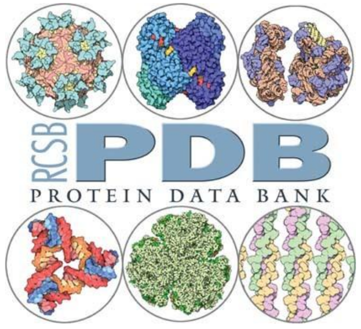
Scientific Method



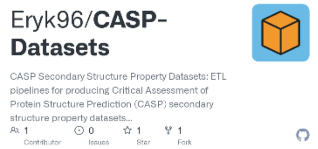
AI for Scientific Discovery

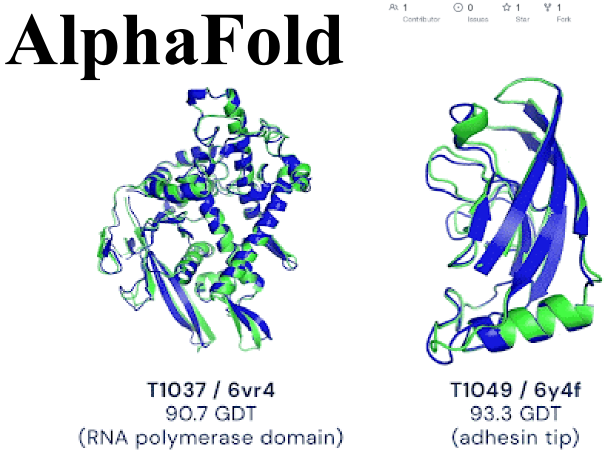
- *AI is now being used to accelerate discoveries in many fields.*

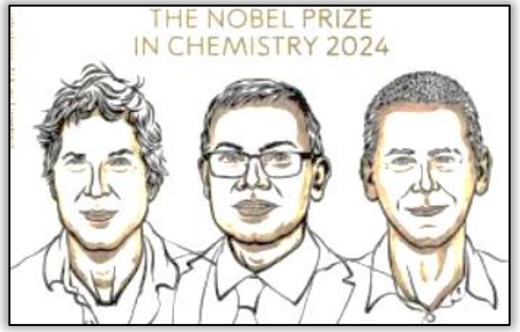




CASP (Critical Assessment of Protein Structure Prediction)







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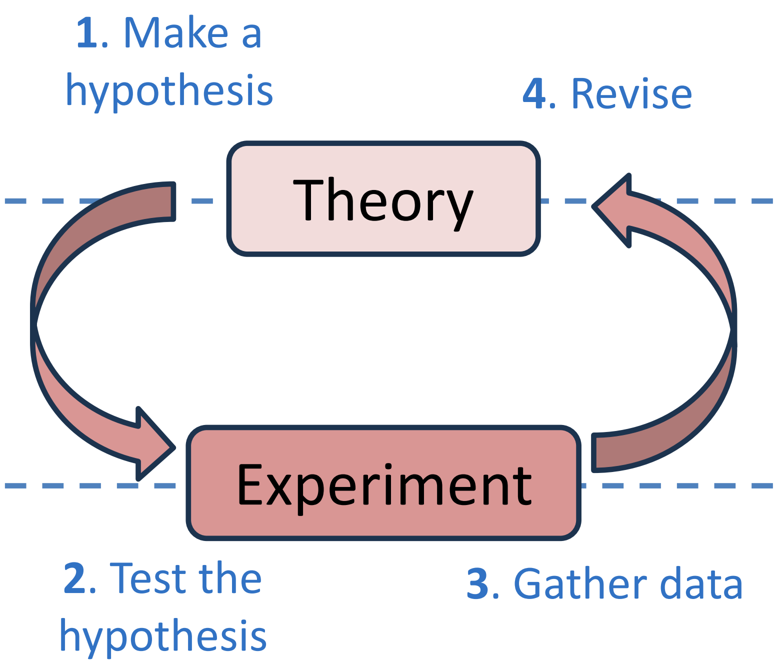
Scientific Method



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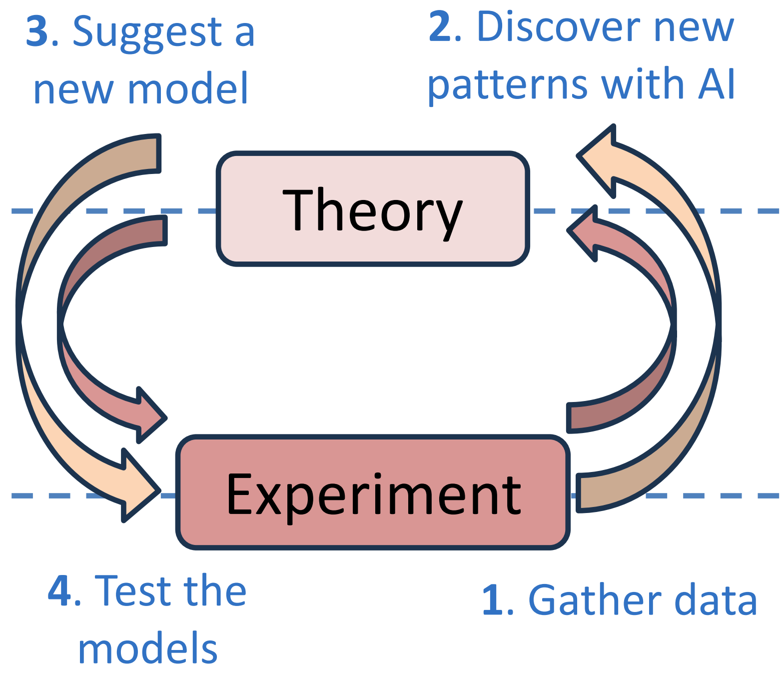
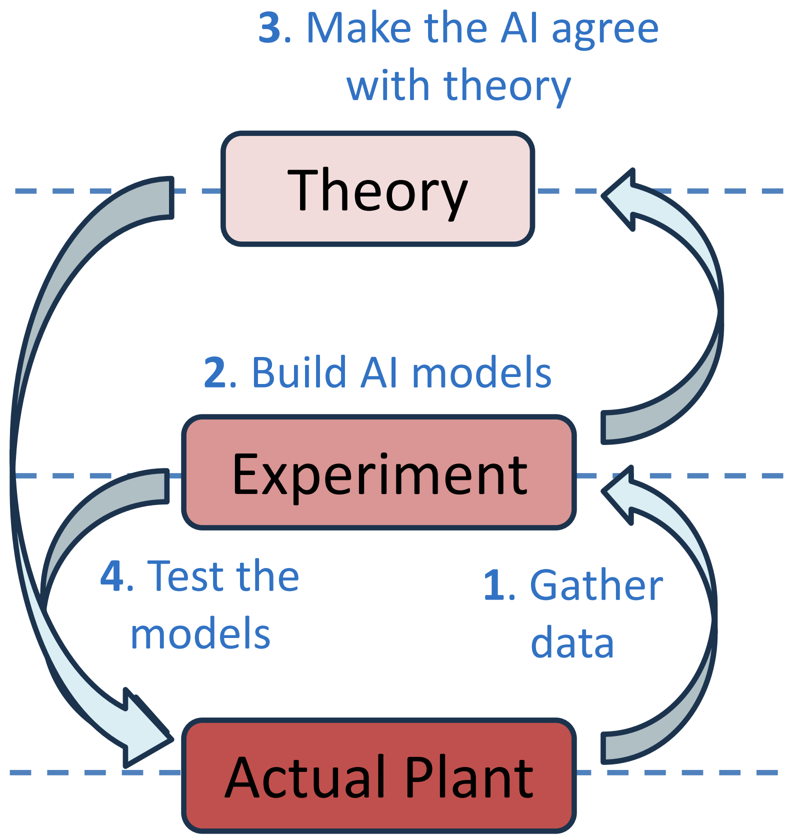


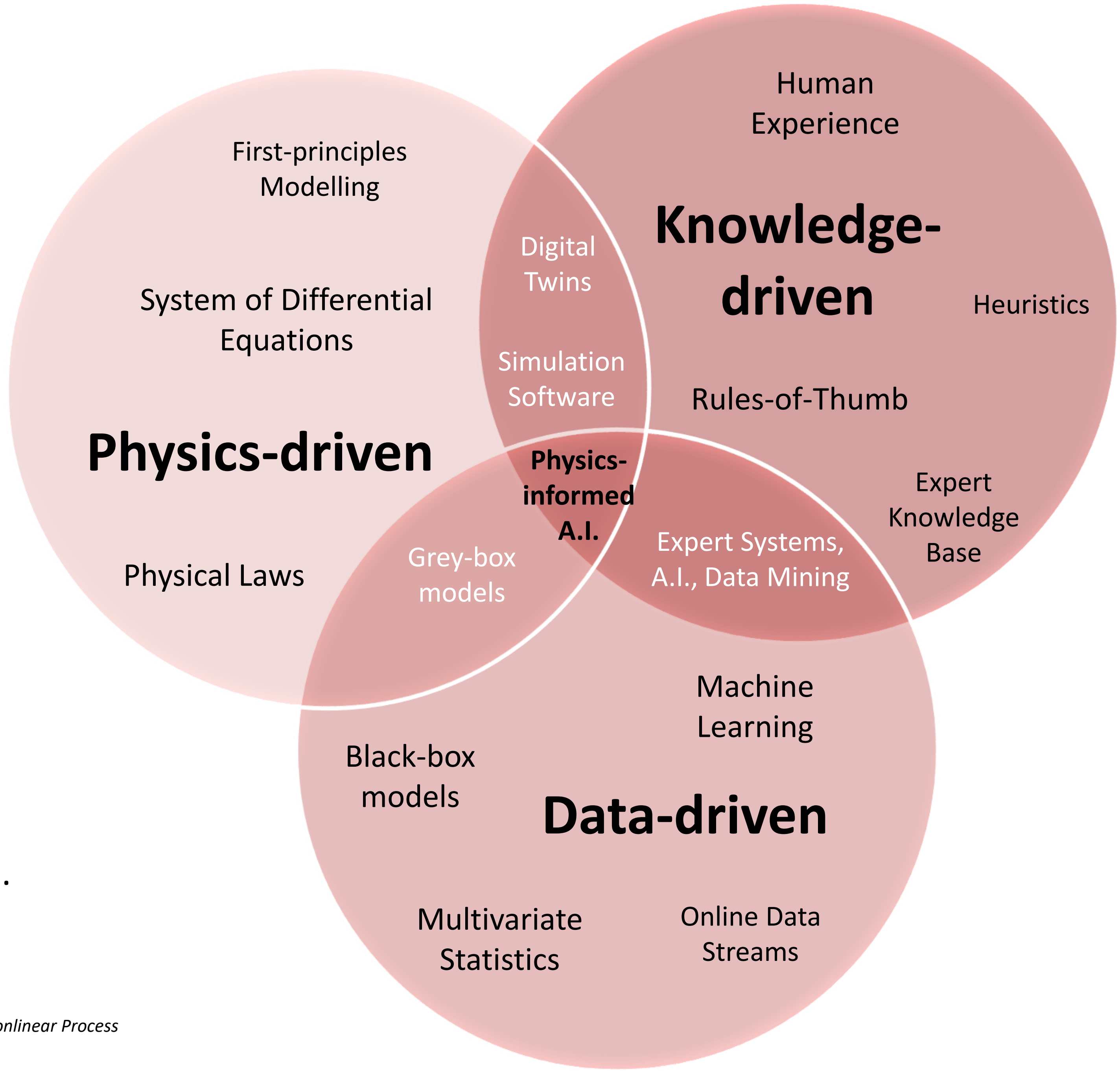
AI for Engineering Practice

- *AI models can be deployed with or without theoretical backing.*



*or any point of application

Why use Machine Learning in your Industry?



Three approaches to engineering problems:

1. Physics-driven Methods
2. Knowledge-driven Methods
3. Data-driven Methods

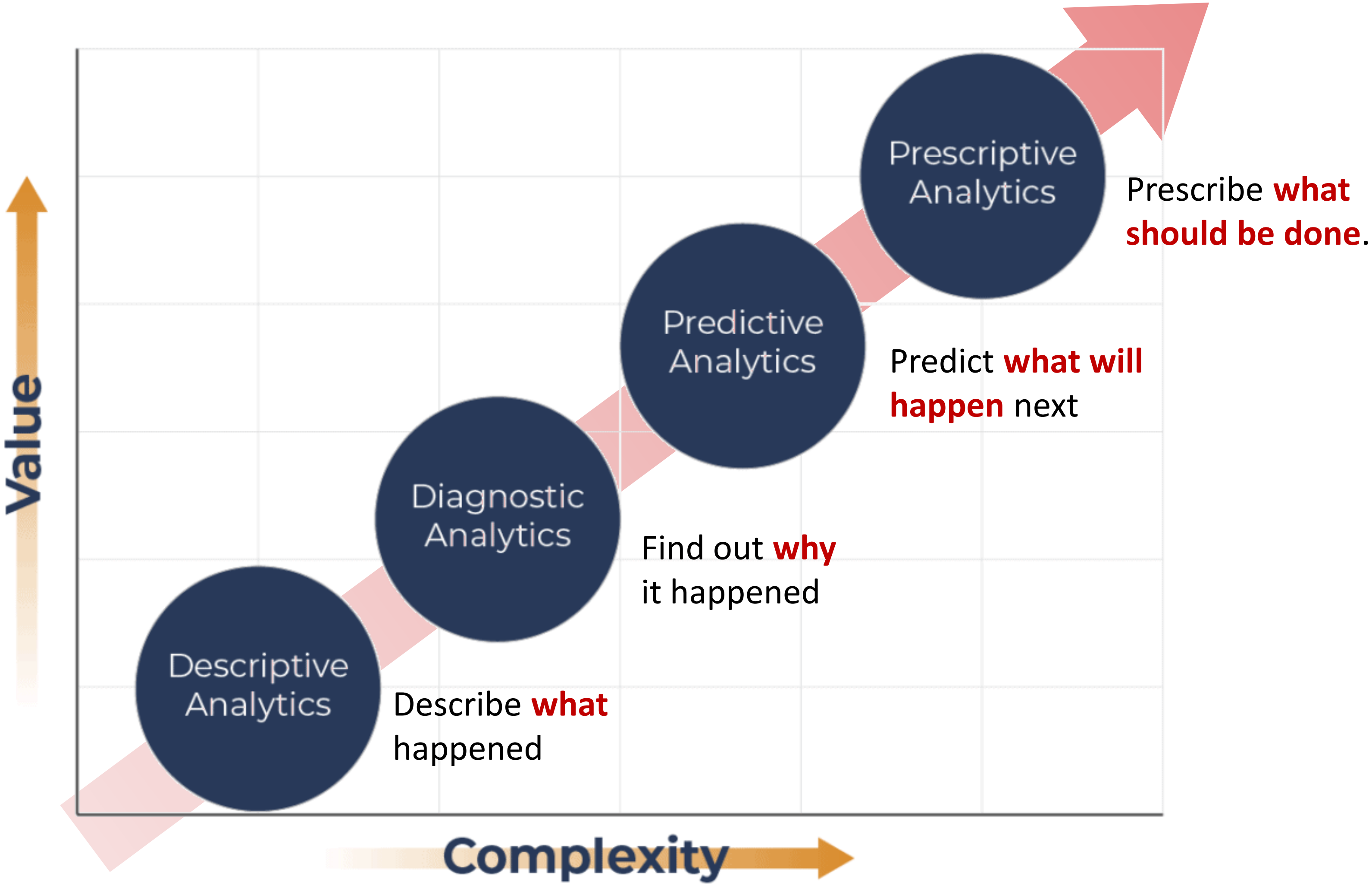
Machine learning is a **data-driven approach**.

Reference: Pilario et al. (2020), *A Review of Kernel Methods for Feature Extraction in Nonlinear Process Monitoring.* MDPI: Processes, https://doi.org/10.3390/pr8010024

How to turn data into decisions?

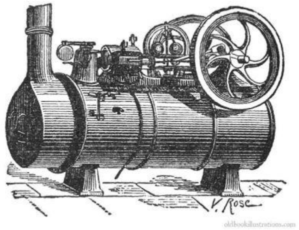
Source: https://iterationinsights.com/article/where-to-start-with-the-4-types-of-analytics/

- Applying machine learning to your data is not enough.
- Don't just let your data speak, let it change the way you do things. **The goal is prescriptive analytics!**
- Getting through each stage of analytics requires more and more effort, but also **more returns**.



"AI is the new electricity."

- Andrew Ng, *2017*



Steam Engine
(1700s)

Light Bulb
(Edison)
1879

Only 3% of US households had electricity
1900

50% of US households now had electricity
1920

Electricity

1820 1850 1880 1910

The Between Times
(Agrawal et al, 2022)

AlexNet (2012)

ChatGPT (2022)

A.I.

2010 2040 2070 2100

The Between Times?
(Agrawal et al, 2022)

Source:
https://www.wsj.com/video/andrew-ng-ai-is-the-new-electricity/56CF4056-4324-4AD2-AD2C-93CD5D32610A
Agrawal, Gans, Goldfarb (2022). Power and Prediction: The Disruptive Economics of AI.

Will machines replace us?

Source: https://binaryverseai.com/llm-hallucinations-explained/

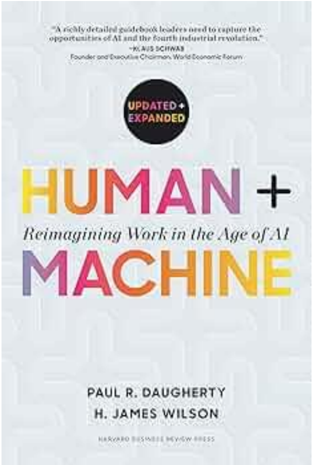
AI is not perfect. So are humans.

Image classification?

Large Language Models (LLM) Hallucinations

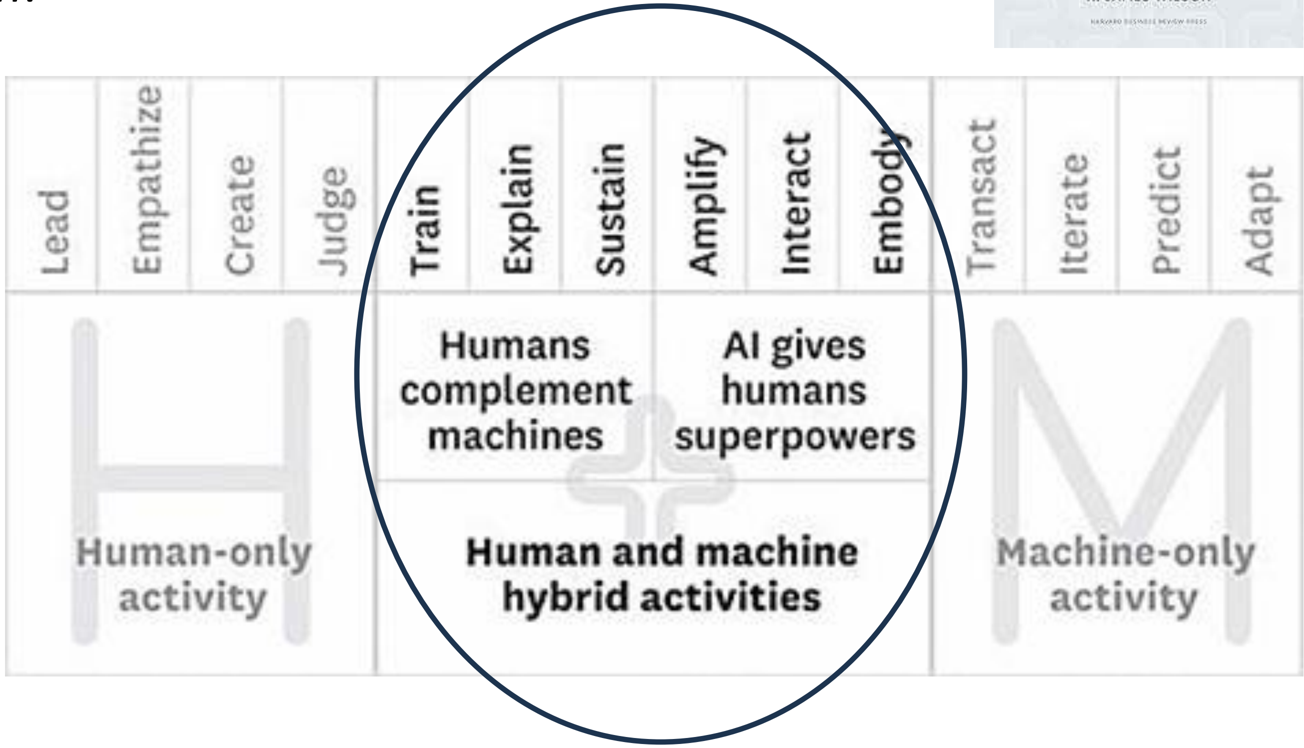
Will machines replace us?

Source: Daugherty and Wilson (2018). Human+Machine: Reimagining Work in the Age of AI.



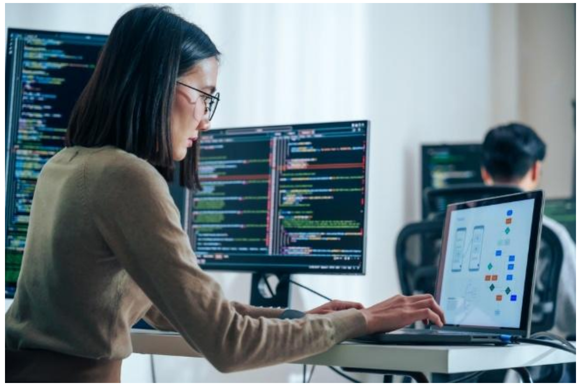
Human-AI collaboration is the next frontier.

An effective AI makes humans *more human*.



New AI Careers in the Job Market

Source: Goldfarb, Agrawal, Gans (2022). Power and Prediction: The Disruptive Economics of AI.



AI Builders

designing, creating, and deploying AI models and applications.

- AI Engineers
- Data Scientists
- Robotics Engineers
- Prompt Engineers

AI Maintainers

ensure the smooth, ongoing operation, efficiency, and reliability of AI systems post-deployment.

- Data Engineers
- Data Scientists
- AI Operators
- AI Maintenance

AI Explainers

building trust, meeting regulatory requirements, and managing the human-AI interaction

- AI Content Reviewers
- Ethics and Compliance Specialists
- Data Storytellers
- Human-AI Collaboration

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  - **Types of Learning Problems**

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Types of Learning Problems

A simple example…

Supervised Learning

These are images of dogs.

These are images of cars.

Now, what is this an image of?

Unsupervised Learning

Here are some images…

Is there an image that does not belong?

Are there images with similar patterns?

Types of Learning Problems

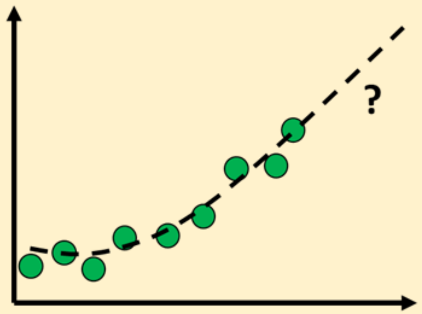
Supervised Learning

Learn a mapping or a function:

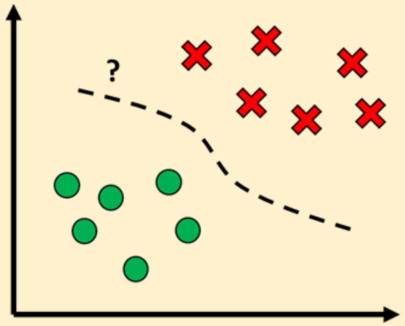
$$y = f(x)$$

from inputs (x) to outputs (y) , *given a labelled set of input-output examples* (or).

Regression



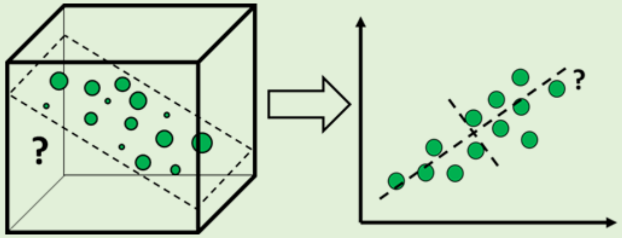
Classification



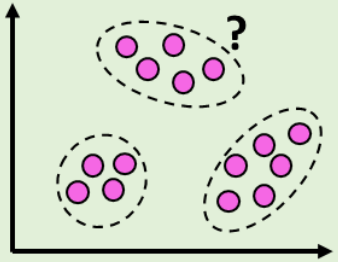
Unsupervised Learning

Discover *patterns or structure* from a data set ( ) without any label information.

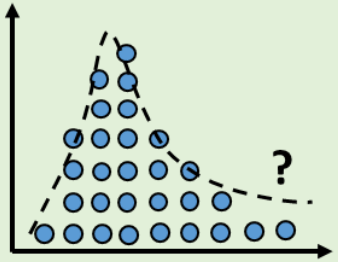
Dimensionality Reduction



Clustering



Density Estimation

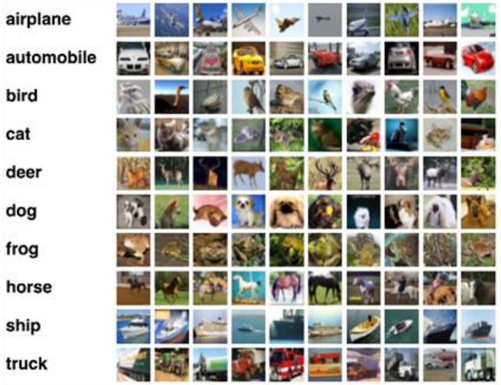


Types of Learning Problems

Semi-Supervised Learning

Goal: Make a computer learn from both labelled and unlabelled data.

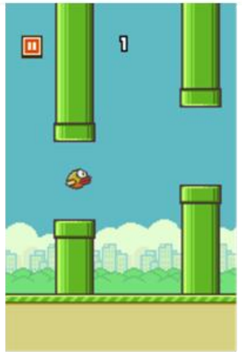
Labelled Data

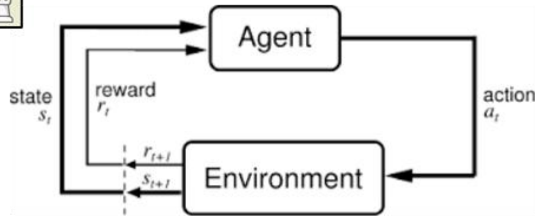


Unlabelled Data

Reinforcement Learning

Goal: Make a computer learn by letting it interact with the environment.





Supervised Learning

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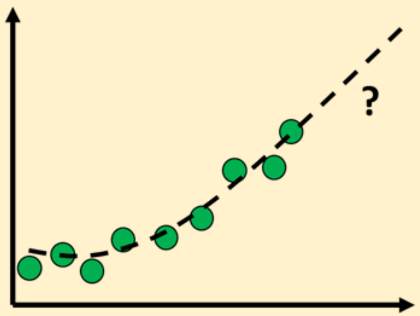
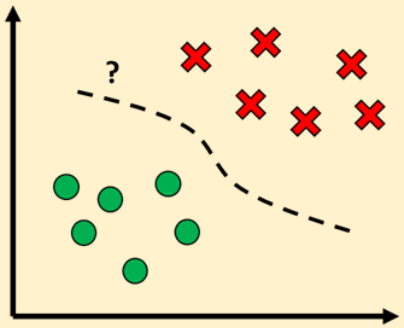
Regression



- **Given:** Training Data $\{\boldsymbol{x}_i, y_i\}_{i=1,2\ldots,N}$
- Target y_i is a **continuous** variable.
- <u>Examples:</u>
  - Forecasting future stock price
  - Forecasting energy resources
  - Prediction of key performance indicators
  - Predicting the properties of molecules based on their structure
  - Predicting the environmental impact of pollutants

Classification

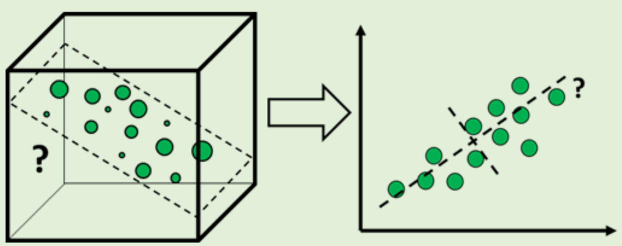


- **Given:** Training Data $\{\boldsymbol{x}_i, y_i\}_{i=1,2\ldots,N}$
- Target y_i is a **categorical** variable.
- <u>Examples:</u>
  - Classifying objects in images
  - Classifying chest X-ray images into COVID positive/negative
  - Handwritten digits recognition
  - Filter e-mails into spam/not spam
  - Classify critical equipment as to healthy or faulty
  - Activity recognition from wearable devices

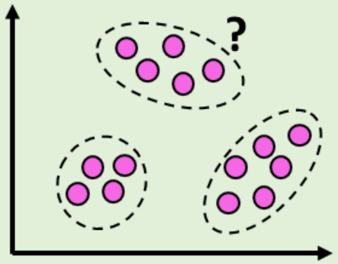
Unsupervised Learning

Discover *patterns or structure* from a data set ( ) without any label information.

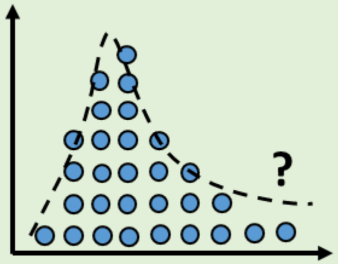
Dimensionality Reduction



Clustering



Density Estimation



Dimensionality Reduction

- **Given:** Data $\{\boldsymbol{x}_i\}_{i=1,2\ldots,N}$
- **Reduce features** while retaining the important information.
- <u>Examples:</u>
  - Feature Engineering
  - Image compression
  - Filtering noise from signals
  - Source separation in audio
  - Data visualization

Clustering

- **Given:** Data $\{\boldsymbol{x}_i\}_{i=1,2\ldots,N}$
- **Group** similar data points together.
- <u>Examples:</u>
  - Customer segmentation
  - Recommendation systems
  - Identifying fake news
  - Clustering documents, tweets, posts

Density Estimation

- **Given:** Data $\{\boldsymbol{x}_i\}_{i=1,2\ldots,N}$
- **Estimate** the distribution of the data.
- <u>Examples:</u>
  - Anomaly Detection
  - Novelty Detection
  - Generative Models
  - Finding distribution modes
  - Spatio-temporal analytics

Can you identify the type of learning problem?

Regression, Classification, Dimensionality Reduction, Clustering, Density Estimation

Example 1

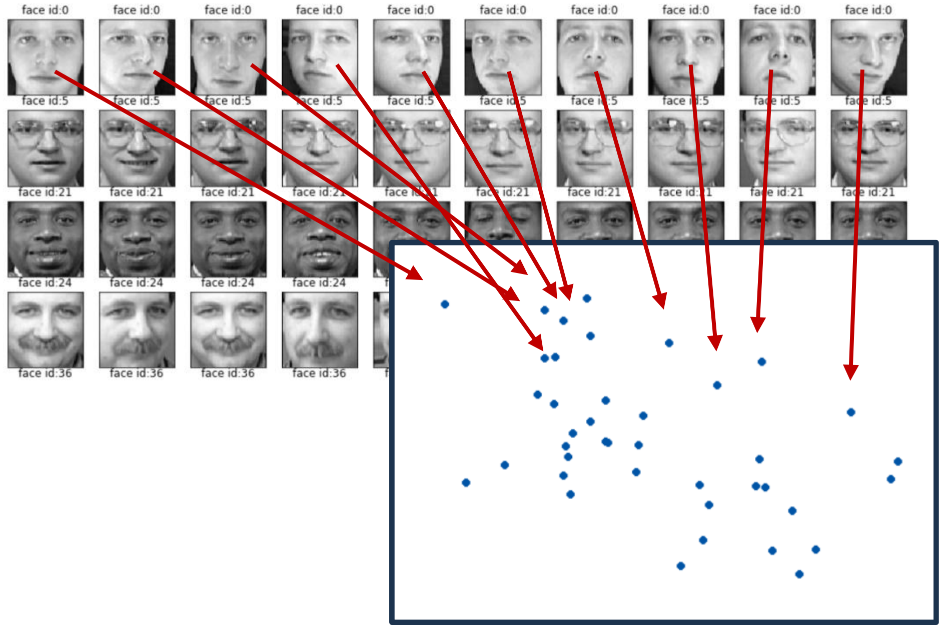
Given the weight of the car, its model year, and horsepower, predict its mileage in miles per gallon (mpg).

car_weight	model_year	horsepower	mileage
1522 kg	2020	150	18 mpg
1930 kg	2017	185	16 mpg
1321 kg	2018	200	21 mpg
2128 kg	2019	168	?
2498 kg	2018	170	15 mpg
1882 kg	2021	155	17 mpg
1956 kg	2019	190	?
1672 kg	2017	182	18 mpg

Answer: Regression

Example 2

Given images of faces with varying poses and expressions, *map* each image onto a 2D point so that similar-looking images are closer together on the map.



Answer: Dimensionality Reduction

Can you identify the type of learning problem?

Regression, Classification, Dimensionality Reduction, Clustering, Density Estimation

Example 3

Given a tweet, predict whether the sentiment is positive, negative, or neutral.

Tweet	Sentiment
I'm in pain…	Negative
Manifesting a promotion this year!	Positive
It's 2AM. Who's awake?	Neutral
Heavy traffic at EDSA	Negative
Family dinner… So full!	Positive
Spoiler alert: RIP Tony Stark	?
Tesla sucks!	?
It's a boy!	Positive

Answer: Classification

Example 4

Given student grades in 5 subjects: Math, Chemistry, Physics, English, and Reading, group the students with similar competencies.

Student	Math	Chemistry	Physics	English	Reading
1	81	85	88	94	92
2	95	80	94	93	85
3	92	94	89	81	80
4	94	83	90	91	84
5	88	84	90	97	95
6	90	93	88	85	82
7	92	94	91	87	81
8	87	82	85	93	94

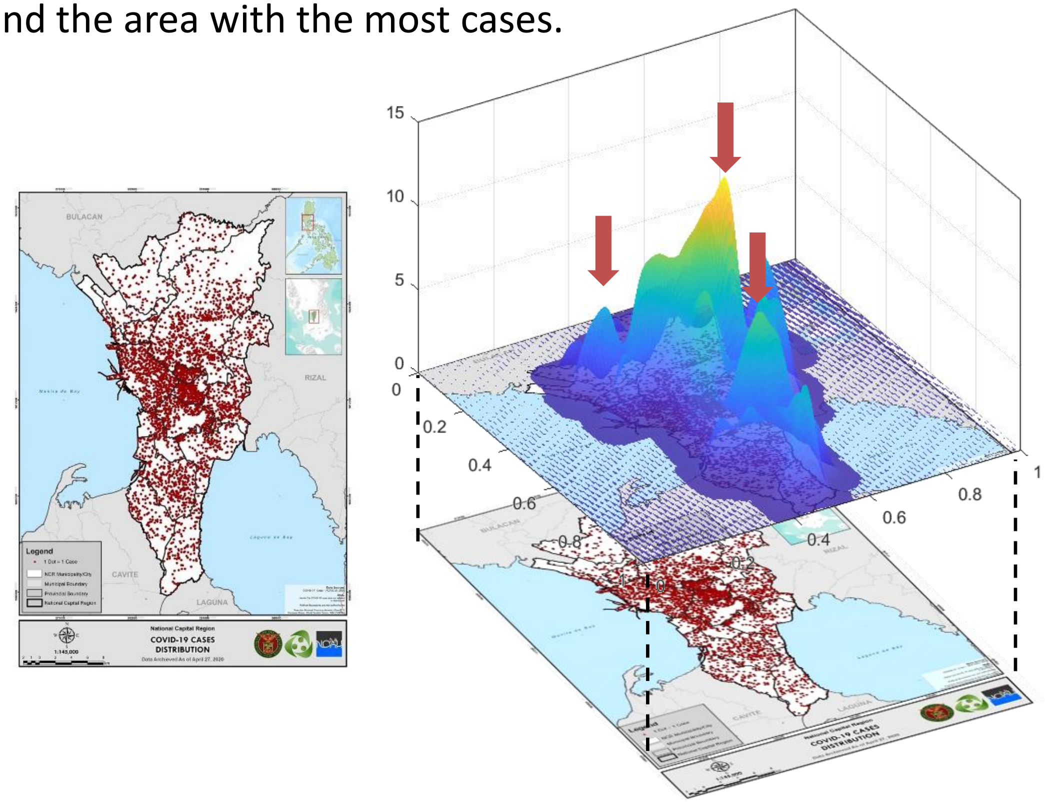
Answer: Clustering

Can you identify the type of learning problem?

Regression, Classification, Dimensionality Reduction, Clustering, Density Estimation

Example 5

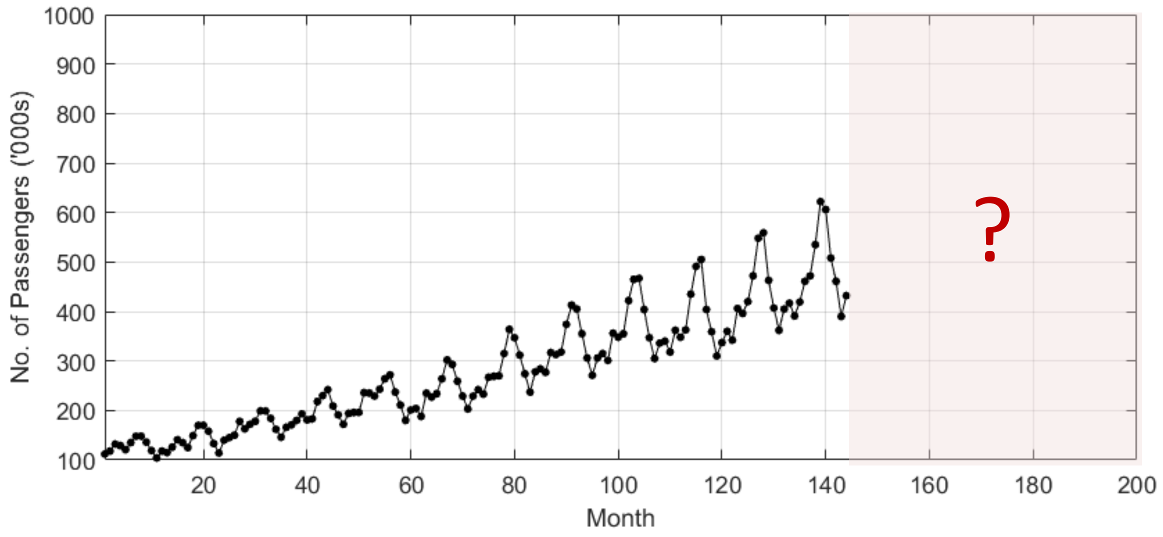
Given the spatial occurrence of Covid cases in Metro Manila, find the area with the most cases.



Answer: Density Estimation

Example 6

Given the number of airline passengers in the previous months, predict the number of passengers for the next few months.



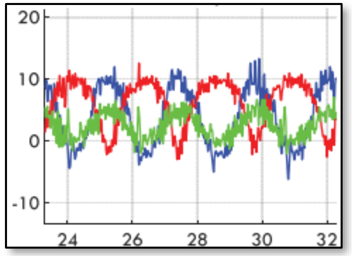
Answer: Regression

Can you identify the type of learning problem?

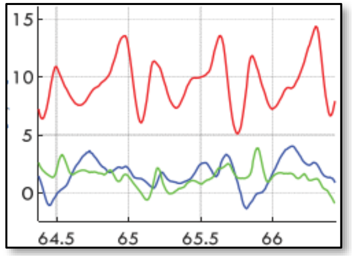
Regression, Classification, Dimensionality Reduction, Clustering, Density Estimation

Example 7

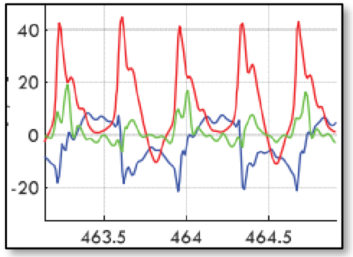
Given smartphone *accelerometer data* from a human doing exercise, predict the kind of exercise being done.



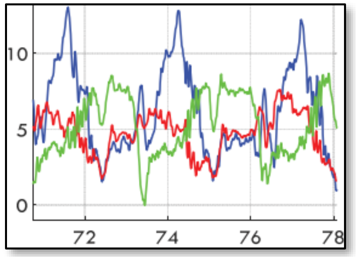
Push-up



Walking



Running



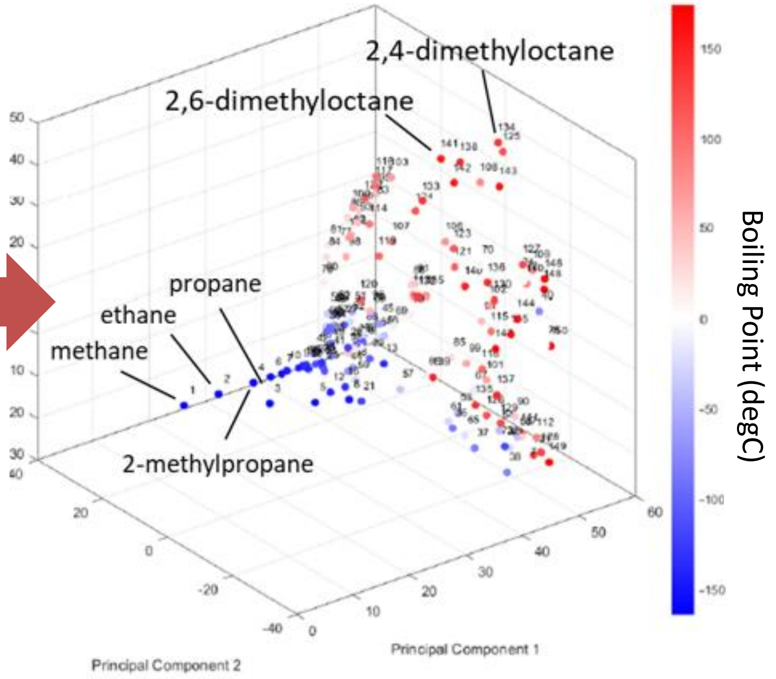
Squats

Answer: Classification

Example 8

Given the structural properties of alkane molecules, *map* them onto 3D space based on their similarities, then *predict* their boiling points.

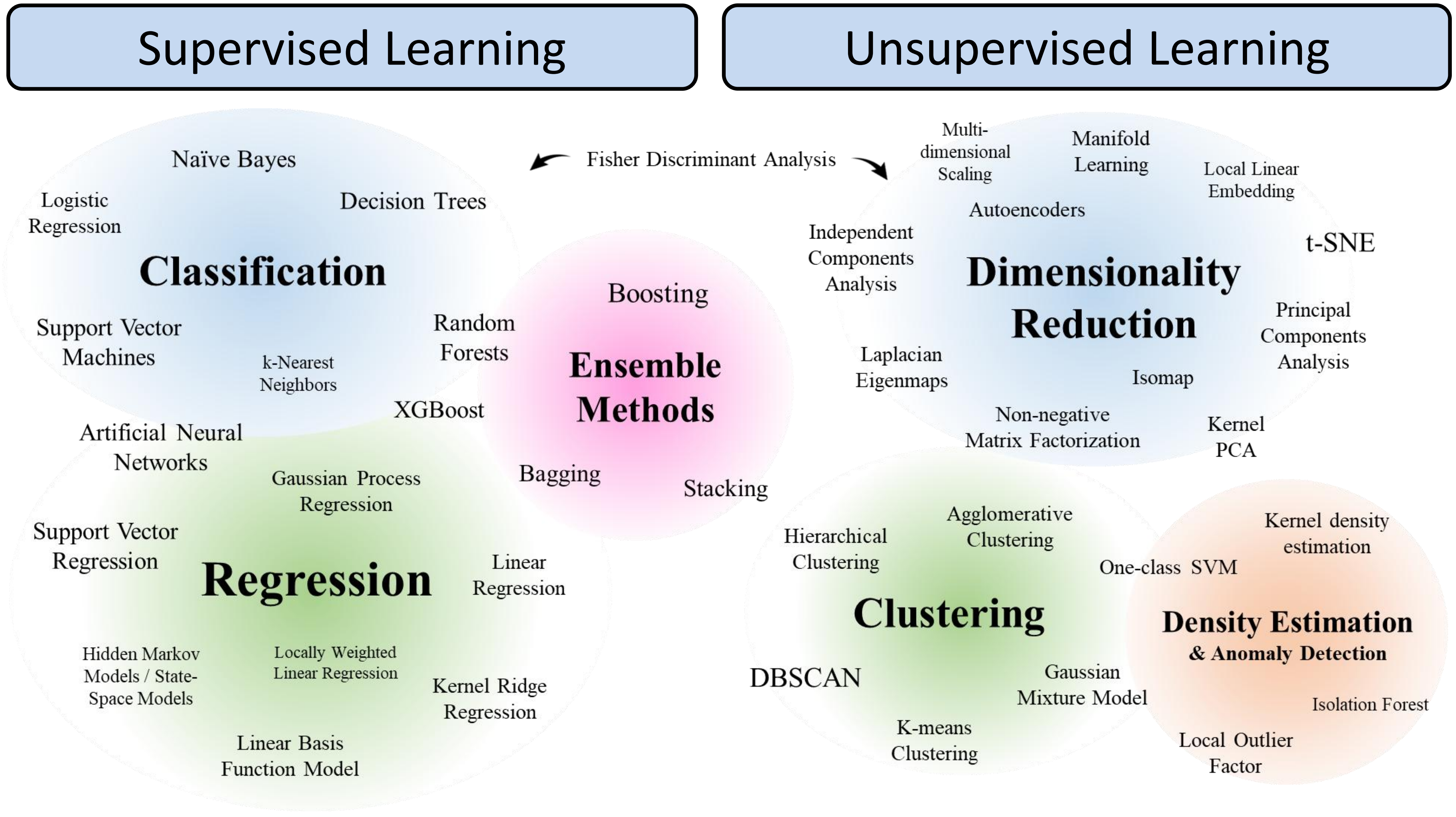
No.	BP	Alkane
1	-164	methane
2	-88.6	ethane
3	-42.1	propane
4	-11.7	2-methylpropane
5	-0.5	butane
6	9.5	2,2-dimethylpropane
7	27.8	2-methylbutane
8	36.1	pentane
9	49.7	2,2-dimethylbutane
10	58	2,3-dimethylbutane
11	60.3	2-methylpentane
12	63.3	3-methylpentane
13	69	hexane
14	80.9	2,2,3-trimethylbutane
15	79.2	2,2-dimethylpentane
16	86.1	3,3-dimethylpentane
17	89.8	2,3-dimethylpentane
18	80.5	2,4-dimethylpentane
19	90	2-methylhexane
20	92	3-methylhexane
21	93.5	3-ethylpentane



Answer: Dimensionality Reduction + Regression

Source: https://ieeexplore.ieee.org/document/8472444

Machine Learning Methods



Reference: Pilario et al. (2020), *A Review of Kernel Methods for Feature Extraction in Nonlinear Process Monitoring.* MDPI: Processes, https://doi.org/10.3390/pr8010024

Outline

- What is Machine Learning?
  - Why only now?
  - Why use ML in your industry?
  - How to turn data into decisions?
  - Will machines replace us?
  - Types of Learning Problems

- **Intro to the Course (AI 221)**
  - Course Delivery
  - Course Content
  - Course Requirements
  - Software

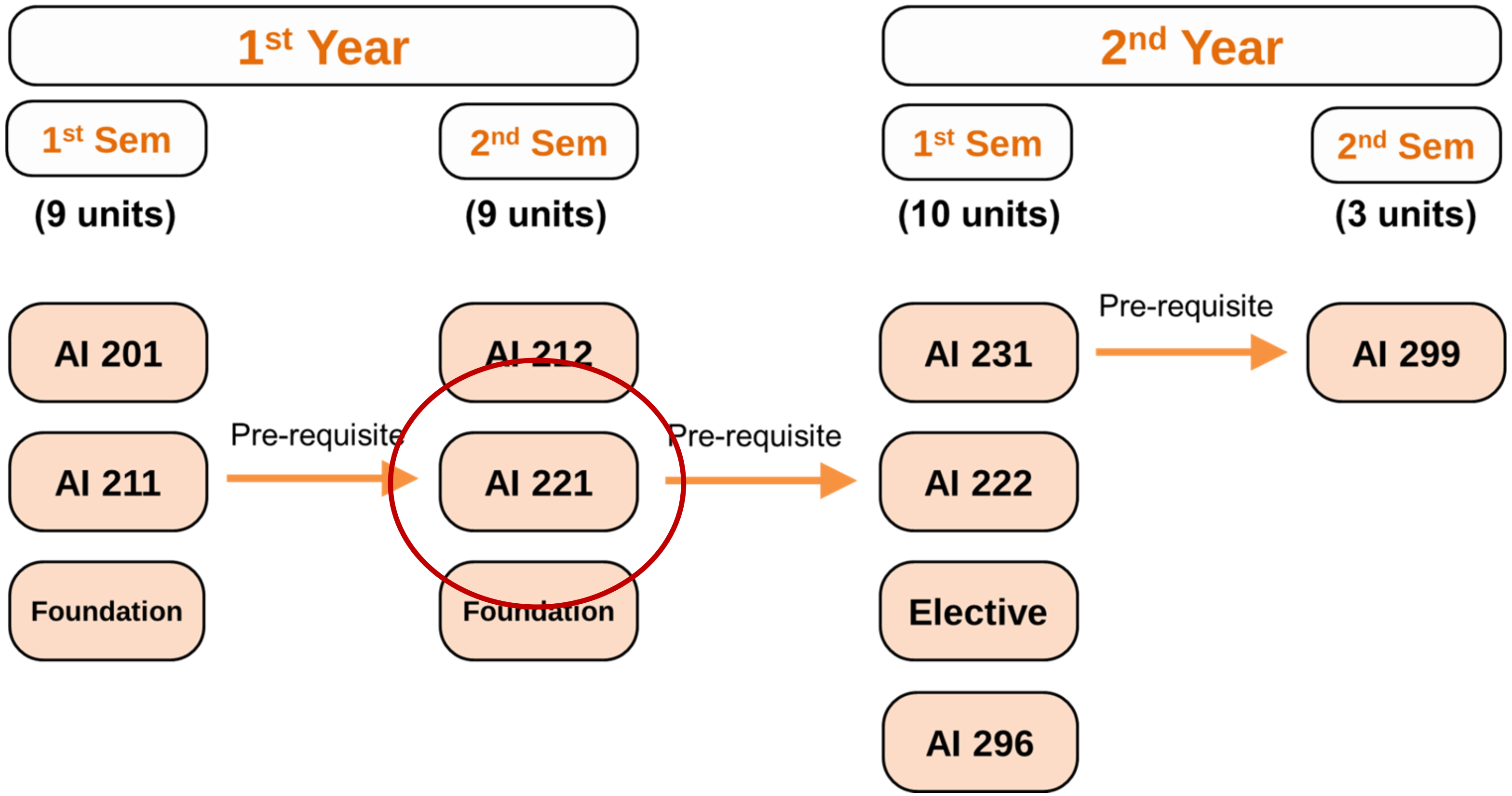
Introduction to the Course

COURSE NUMBER:	AI 221
COURSE TITLE:	Classical Machine Learning
COURSE DESCRIPTION:	Linear Models. Kernel Methods. Neural Networks. Trees. Clustering. Dimensionality Reduction. Feature Engineering. Density Estimation. Ensemble Learning. Gaussian Processes. Bayesian Methods. Hyperparameter Search. AutoML. Explainability.
COURSE CREDIT:	3 units
	3.0 hours/week
Github:	https://github.com/kspilario/AI221

Introduction to the Course

MEngg in Artificial Intelligence

Total Units: 31 Units

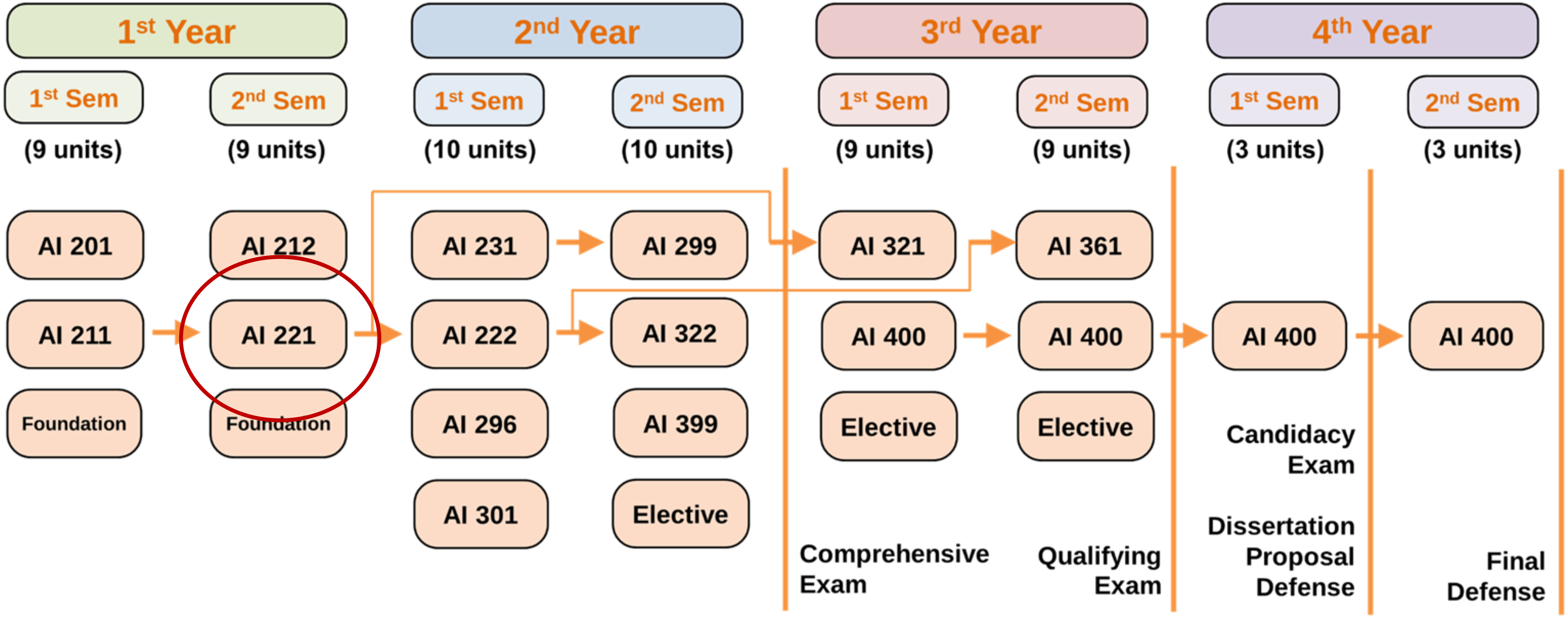


Introduction to the Course

PhD in Artificial Intelligence

Option A

Total Units:	62 Units
Requirement:	2 Publications



AI 221 Course Delivery

- **Meeting:** Every Wednesday, online.

Sunday	Monday	Tuesday	**Wednesday**	**Thursday**	Friday	Saturday
			6-9 PM	6-9 PM		

- **Course Requirements:**

Requirement	% of Final Grade	Mode
• Team Project • Oral Presentation (40%) • Written Report (50%) • Graphical Abstract (10%)	**40%**	By groups of 1 to 3 members only, Face-to-face
• Machine Exercises	**40%**	By group (1-3), Take-home
• Journal Critique	**20%**	Individual, Take-home

- **Grading System:**

[92,100]	[88,92)	[84,88)	[80,84)	[76,80)	[72,76)	[68,72)	[64,68)	[60,64)	[0,60)
1.00	**1.25**	**1.50**	**1.75**	**2.00**	**2.25**	**2.50**	**2.75**	**3.00**	**5.00**

AI 221 Course Content

Jan 21	Week 1.	Introduction to Machine Learning
Jan 28	Week 2.	Exploratory Data Analysis
Feb 4	Week 3.	Linear Regression
Feb 11	Week 4.	Logistic Regression
Feb 18	Week 5.	SVM and Kernel Methods
Feb 25	Week 6.	Gaussian Processes and Bayesian Optimization
Mar 4	Week 7.	Hyper-parameter Tuning and AutoML
Mar 11	Week 8.	Neural Networks
Mar 18	Week 9.	Trees and Ensemble Learning
Mar 25	Week 10.	Linear Dimensionality Reduction
		HOLY WEEK
Apr 8	Week 11.	Nonlinear Dimensionality Reduction
Apr 15	Week 12.	Clustering, Density Estimation, and Anomaly Detection
Apr 22	Week 13.	Machine Learning on Time Series
Apr 29	Week 14.	Explainable AI and Case Studies
May?	Week 15.	***Team Project Presentation***

AI 221 Course Requirements

Team Project (40%)

- A team should have at most 3 members only.
- <u>Aims</u>:
  - Find a problem + data set that requires an ML solution.
  - Solve the problem using the ML methods discussed in class.
  - Present your results to the class.
- NO two teams should have the same problem.
- Grading:
  - Oral Presentation (40%)
  - Written Report (50%)
  - Graphical Abstract (10%)

Machine Exercises (40%)

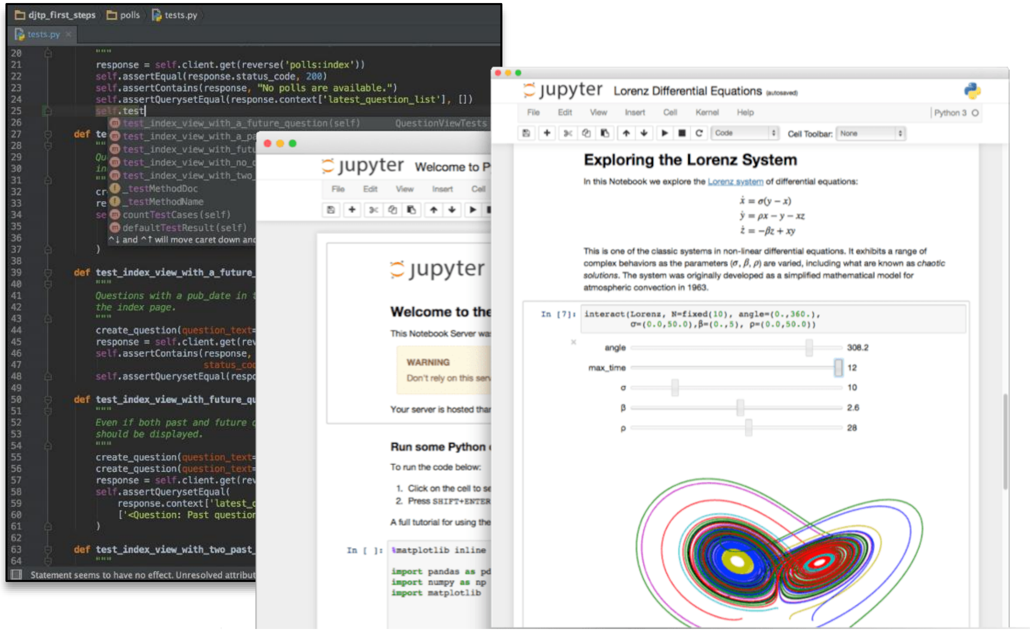
- Mode: Groups of 1-3 members, take-home
- There are **5 MEXes** throughout the semester.
- Submission deadline is **2 weeks** after release.

Journal Critique (20%)

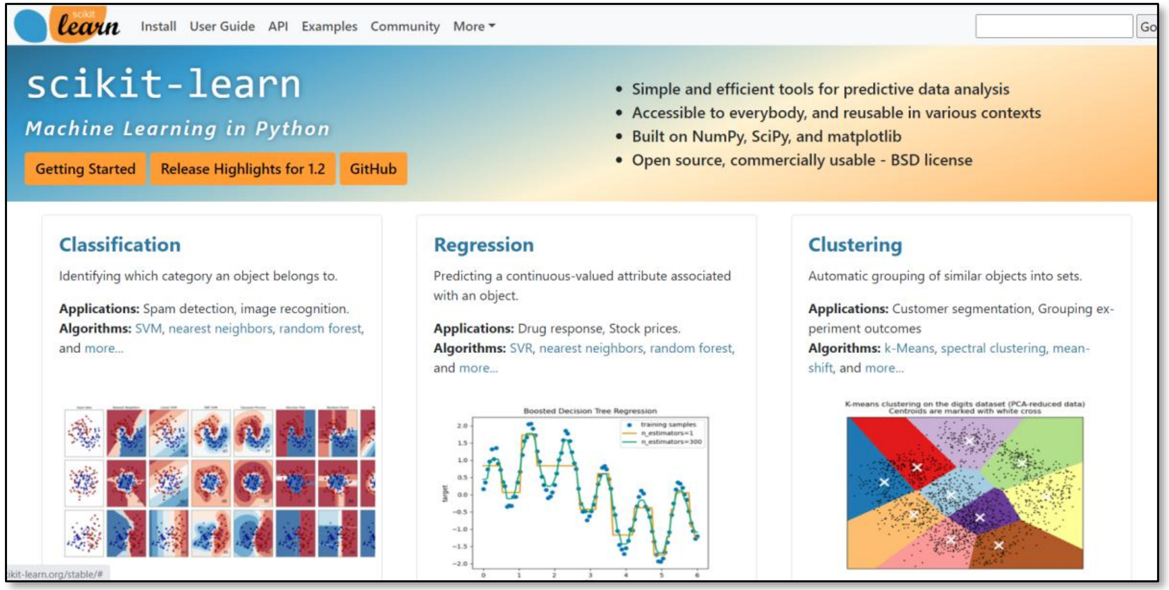
- Mode: Individual, take-home
- Find a paper from a reputable journal or conference proceedings related to your field.
  - Should at least have an impact factor.
  - Should be published in the last 5 years.
- Send me the paper for approval first, then I will send guide questions for you to answer.
- Deadlines: To be announced

AI 221 Required Software

- Anaconda >> Spyder
- Anaconda >> JupyterLab
- Google Colab
- Jupyter Notebook
- PyCharm
- JupyterLite
- Microsoft Excel



- Python 3: https://www.python.org
- Numpy: http://www.numpy.org/
- Scikit-Learn: https://scikit-learn.org/
- Juptyer Lab
  - https://jupyter.org/try-jupyter/lab/
  - https://nbviewer.org/
- MS Excel



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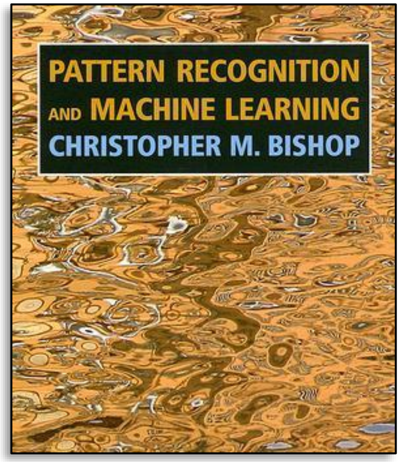
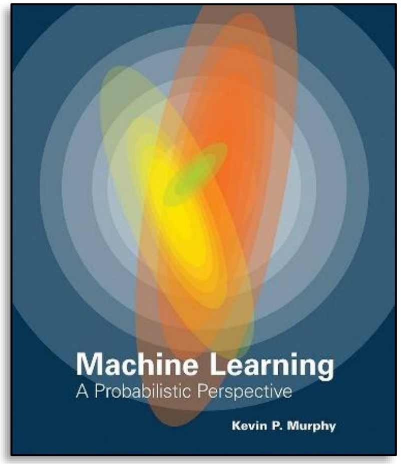
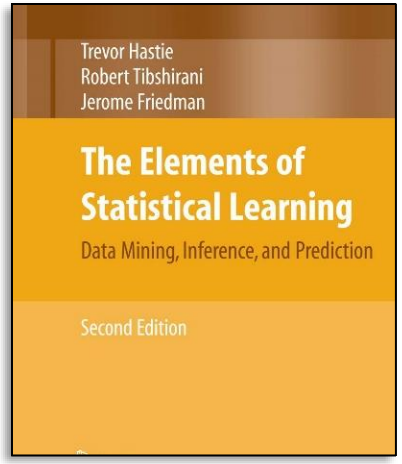
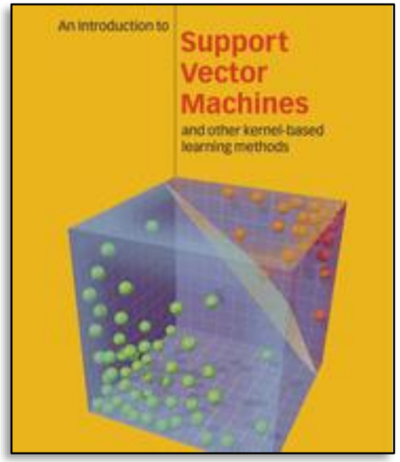
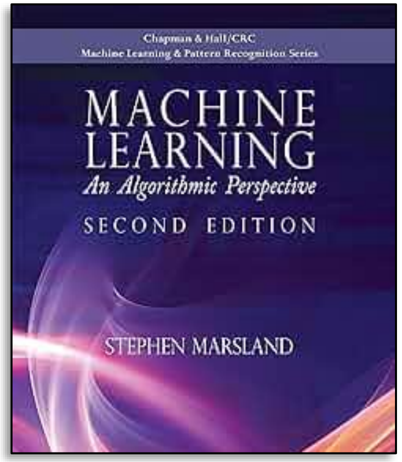
- Murphy, Kevin (2012). *Machine Learning: A Probabilistic Perspective.* MIT Press.
- Hastie et al. (2008). *The Elements of Statistical Learning.* 2nd Ed. Springer.
- Cristianini & Shawe-Taylor (2000). *An Introduction to Support Vector Machines and other kernel-based learning methods.* Cambridge University Press.
- Marsland, Stephen (2014). *Machine Learning: An Algorithmic Perspective*. Chapman and Hall. 2nd Ed.
- Rasmussen and Williams (2006). *Gaussian Processes for Machine Learning.* MIT Press. https://gaussianprocess.org/gpml/
- Geron, Aurelien (2019). *Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow*. O'Reilly Media.
- Journals and Conference Proceedings
- Python API, Sci-kit learn API: https://scikit-learn.org/stable/modules/classes.html
- Online Courses, Youtube Videos, etc.

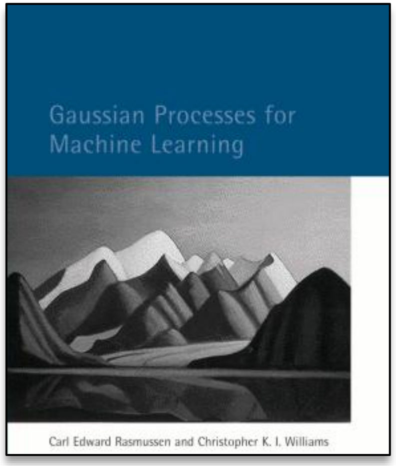


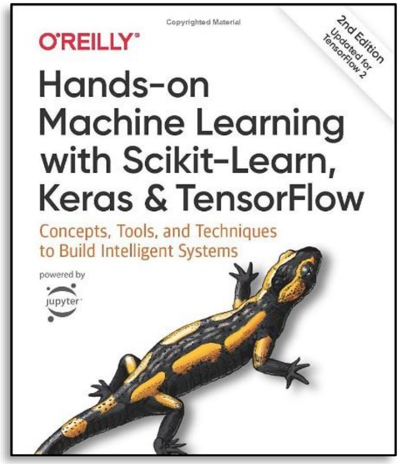












AI 221 Course Instructor

Current Position

Karl Ezra S. Pilario
Professor
Department of Chemical Engineering
University of the Philippines, Diliman

- Process Dynamics & Control
- Programming in MATLAB, Python, Aspen HYSYS
- Numerical Methods in Engineering
- Plant Design and Research
- Machine Learning and Artificial Intelligence

University of the Philippines, Diliman

Research Lab

Head, Process Systems Engineering Laboratory (PSEL)
Department of Chemical Engineering
University of the Philippines - Diliman

Education

- **Bachelor's Degree:**
  Chemical Engineering, *SCL* (**2012**)
  University of the Philippines Diliman
- **Master's Degree:**
  Chemical Engineering (**2015**)
  University of the Philippines Diliman
- **PhD Degree:**
  PhD Energy and Power (**2020**)
  Cranfield University, United Kingdom

Research Interests

- Process Data Analytics
- Process Systems Engineering
- Industrial Process Monitoring and Predictive Maintenance
- Machine Learning for Energy, Water, and Environment
- Cheminformatics and Materials Informatics